



**HKU
Med**

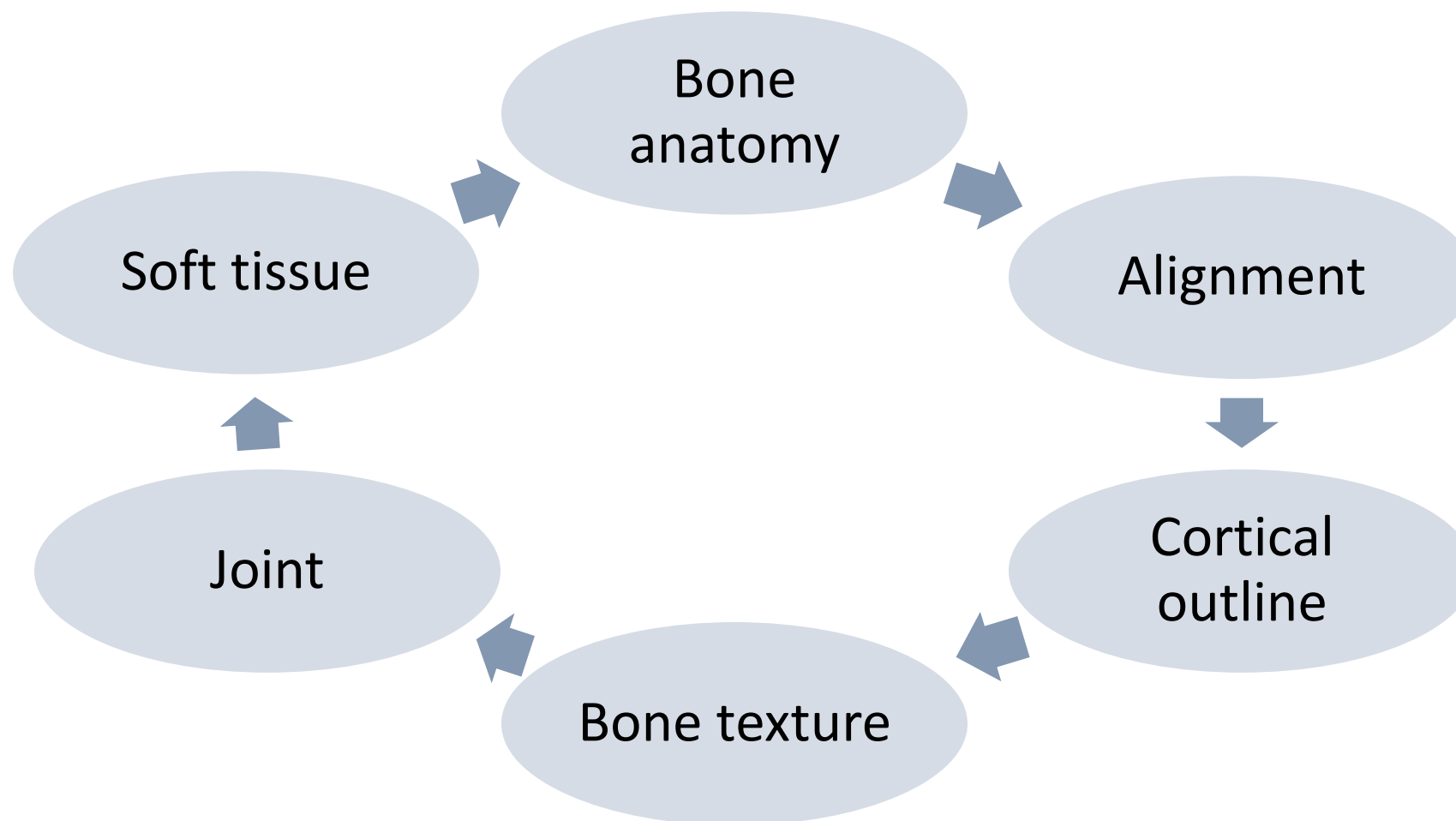
LKS Faculty of Medicine
Department of Diagnostic Radiology
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Musculoskeletal Radiology Junior Clerkship Tutorial

Case Discussion

**If you have not viewed the video
on the e-learning platform
(Moodle), please use the first ½ hour
of the tutorial to do so**

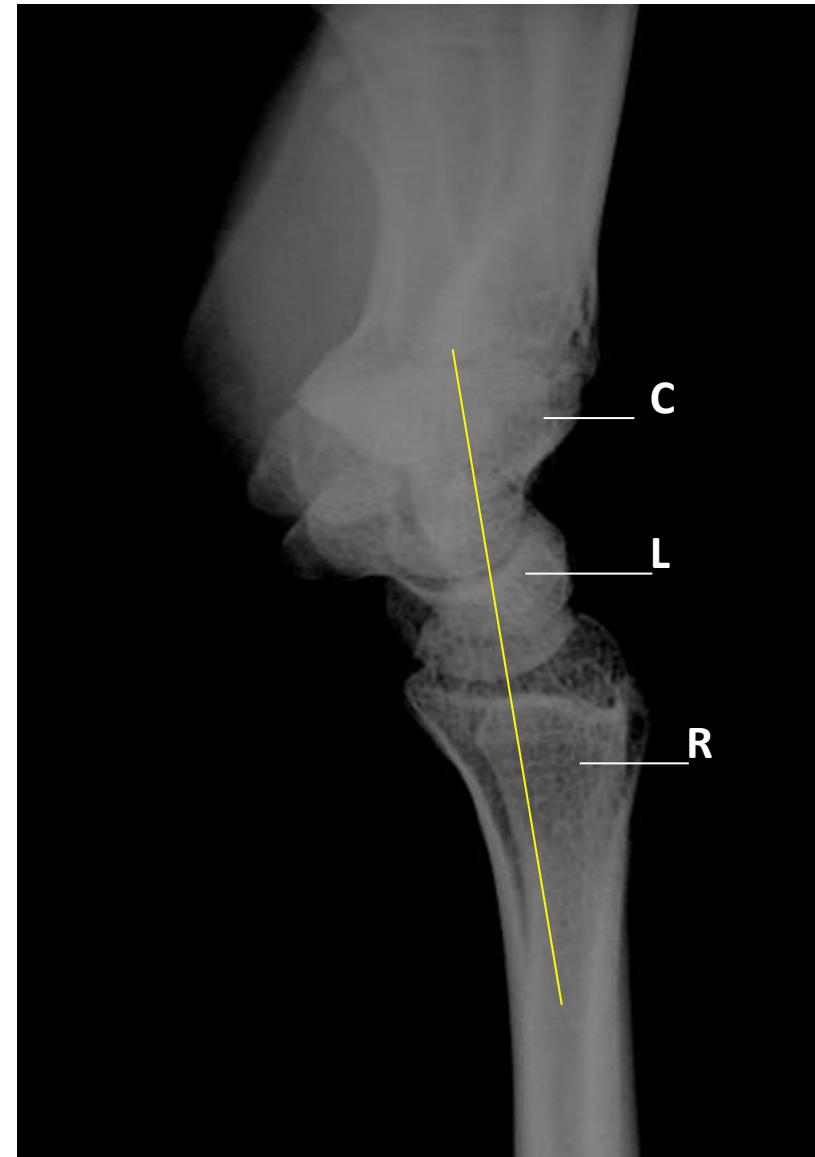
Systematic Approach



1. Patient sustained a fall on an outstretched hand
Is the alignment normal?



Normal x-ray wrist



Perilunate dislocation and distal radial fracture

- disruption of the alignments of the carpal rows
- capitate not sitting in the “cup” of the lunate
- cortical break at the distal radius



C: Capitate

L: Lunate



2. 45-year old male who was the backseat passenger involved in a road traffic accident.
Minimal neurological deficit.

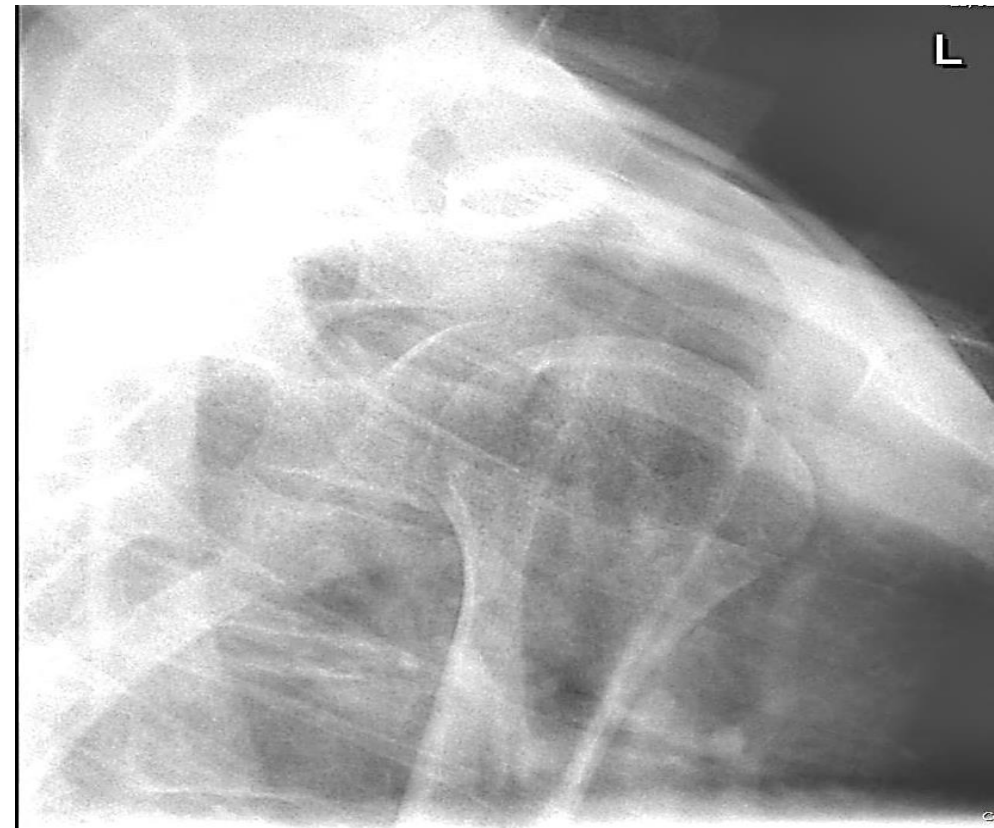
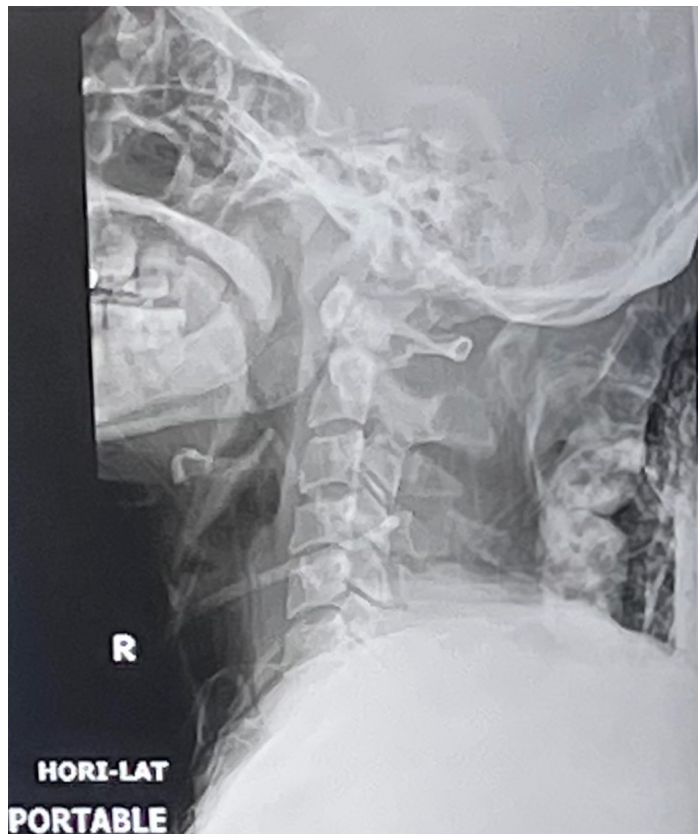


Unilateral facet joint dislocation

- rotated facets on lateral view
- widening of interspinous space



Another road traffic accident-is there any fracture?

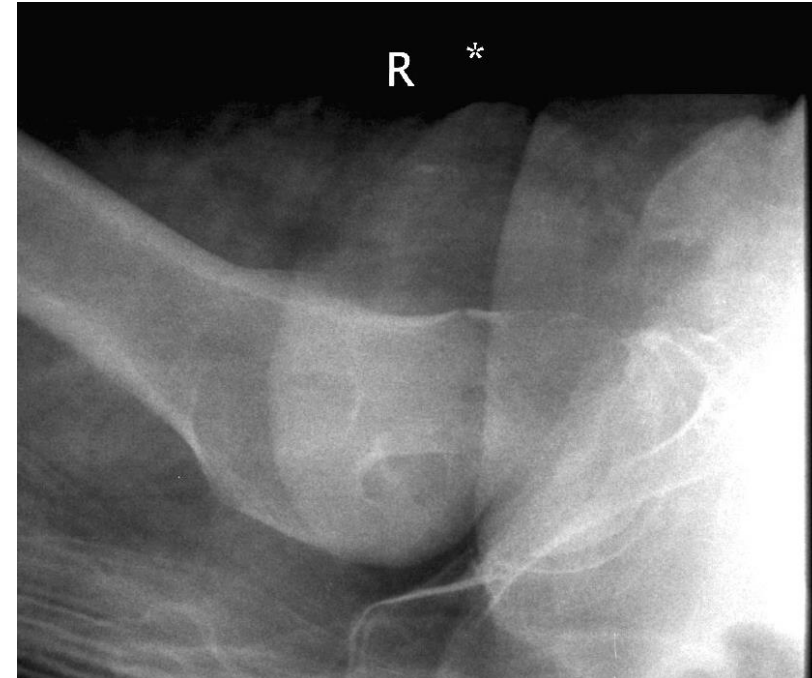


C7 fracture

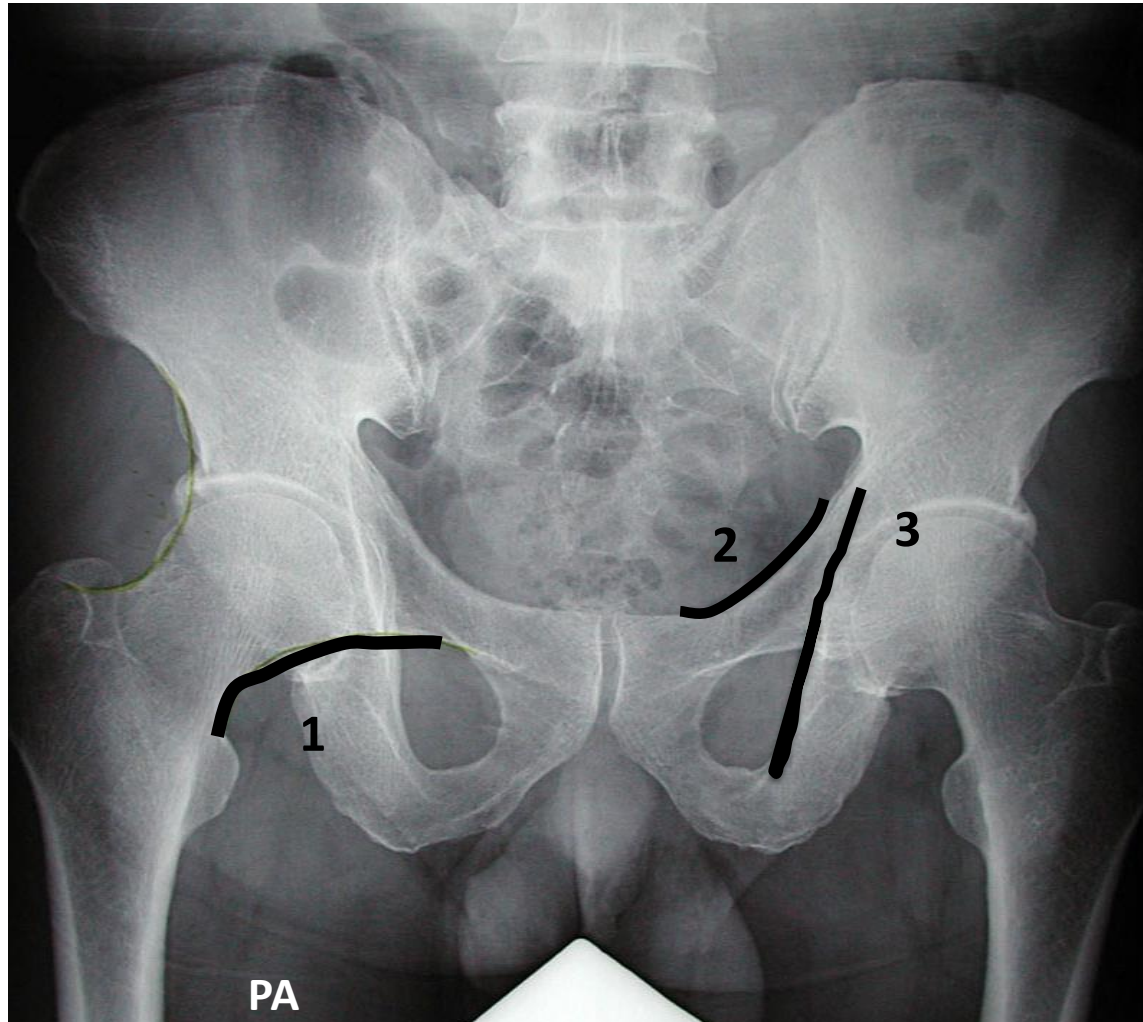
Beware of inadequate visualization of C7/T1



3. 80-year old man sustained minor trauma (slip and fell). Unable to get up and walk. O/E: right hip externally rotated, swelling and decreased ROM.



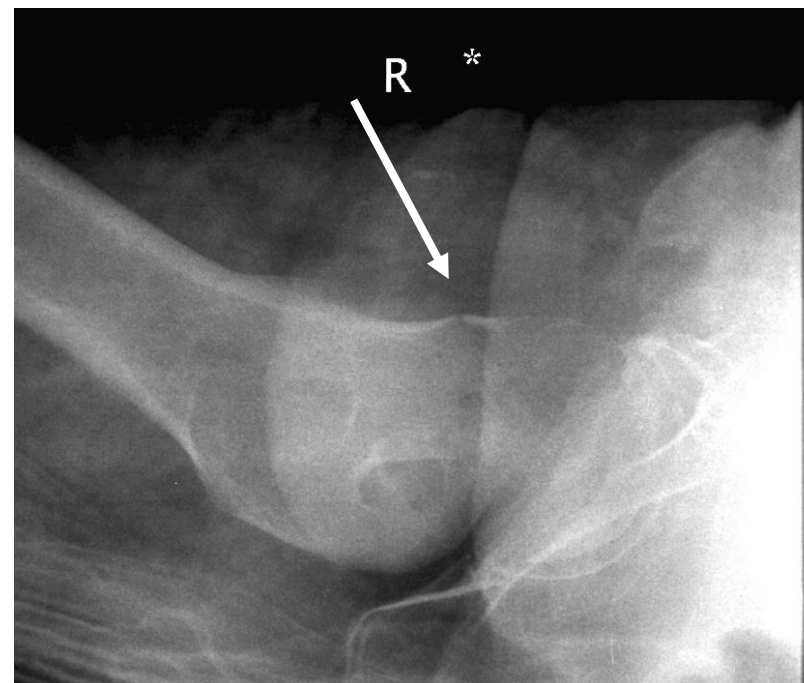
Normal Pelvic X-ray



1. Shenton's line:
Continuous line from
medial femoral neck to
inferior aspect of superior
pubic ramus.
2. Iliopectineal line
3. Ilioischial line

Neck of femur fracture

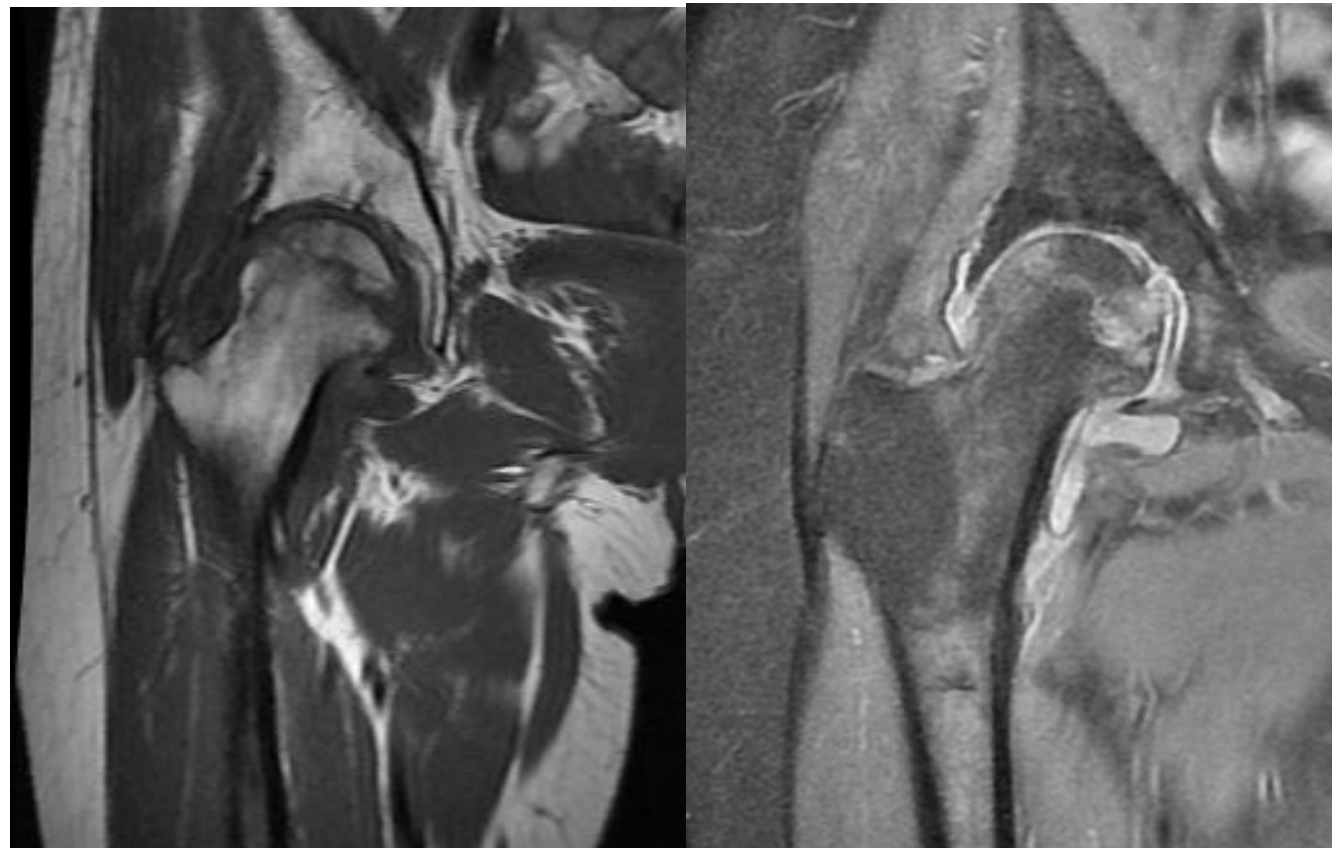
- disruption of Shenton's line
- increased sclerosis due to impaction



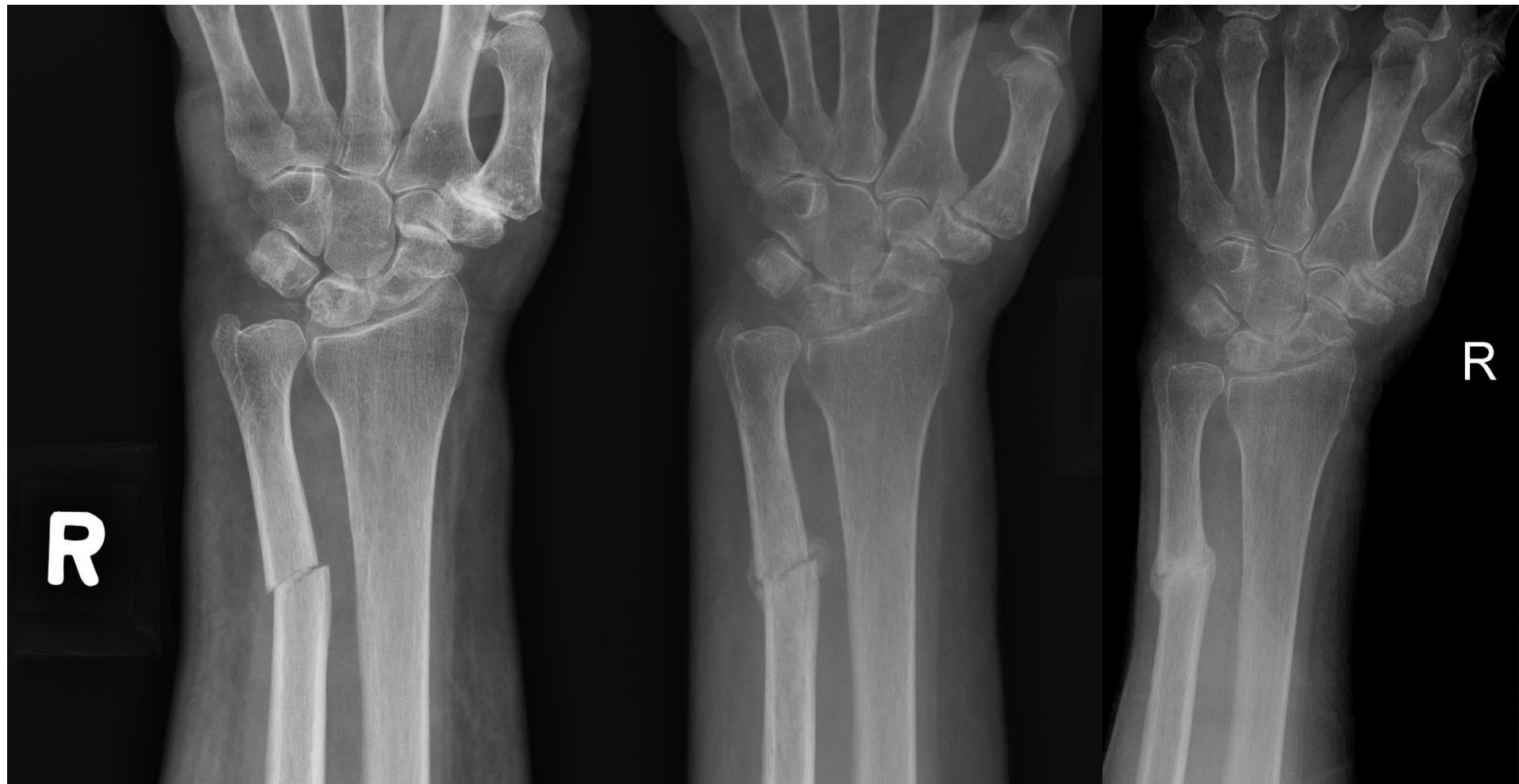
Femoral neck fracture (intracapsular)

- **Most common location:**
subcapital
- **Non-displaced/displaced**
- **Complications:**
 - **Osteonecrosis/AVN** (usually with displaced fractures).
- **Displaced fractures tend to compromise the blood supply to the femoral head.**
- **Other factors:**
 - Increase in intracapsular pressure leading to decreased venous return
 - Reduced arterial perfusion.

Avascular Necrosis



Healing fracture



4. Patient was involved in a road traffic accident.



Lisfranc fracture dislocation

- fractures of base of the metatarsals
- dislocated from the tarsal bones



5. A 30-year old man sustained a deep laceration of the index finger.



Bone texture

Bone lucency

- Osteopenia
- Fracture line
- Destructive process
 - Infection
 - Malignant
- Bone tumour
 - Primary vs. metastasis

Bone sclerosis

- Impacted/healing fracture
- Bone tumour
 - Primary vs. metastasis
- Systemic process
 - Myelofibrosis
 - Sickle cell

Enchondroma

- well-defined margin
- central lucency +/- chondroid calcification
- look for pathological fracture



Bone sclerosis

Sclerotic bone metastases



Paget's disease

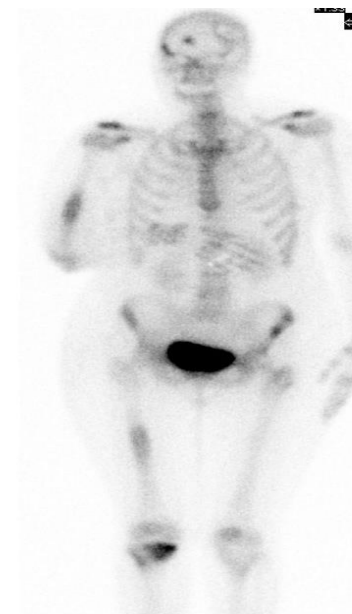


Bone lucency

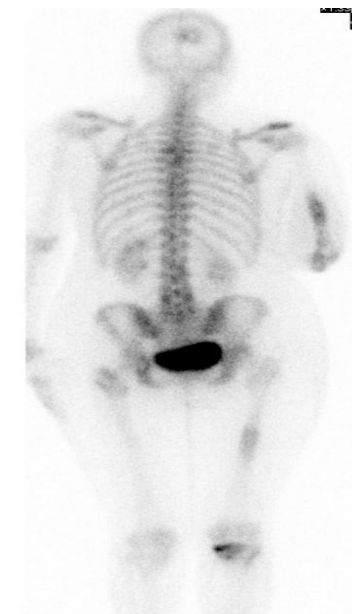
Bone metastasis



Bone scan

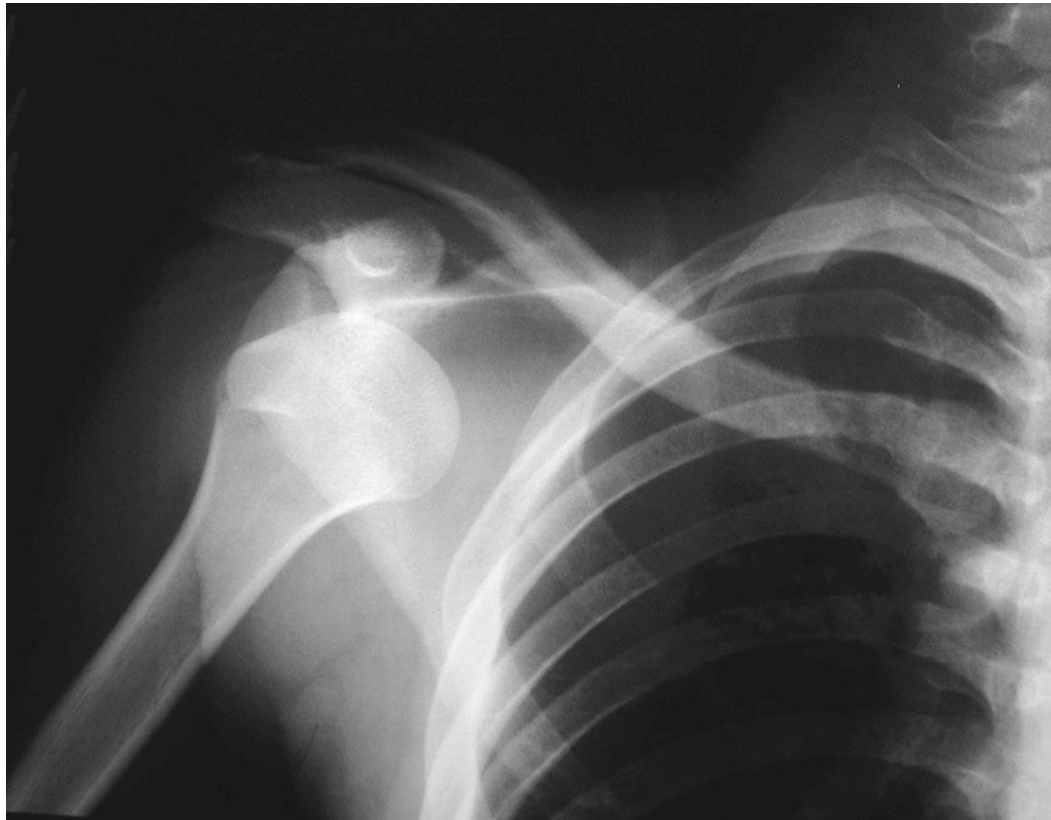


RT ANTERIOR LT



LT POSTERIOR RT

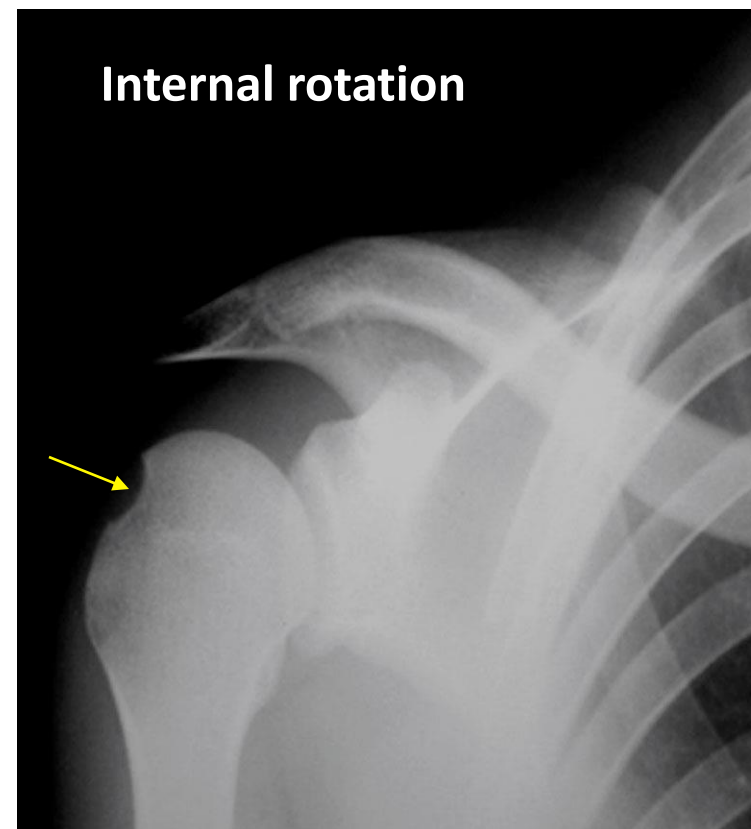
6. 35-year old sustained injury when playing basketball. Unable to move arm and shoulder.



Anterior dislocation of the shoulder



Anterior dislocation: 95%
Associated fracture of the inferior
glenoid margin (Bankart fracture).



Hill-Sachs deformity:
Cortical impaction of superior
posterior aspect of humerus

Case 7

- 40-year old female with several months of low back pain, radiating down the left lower limb.
- O/E: Reduced straight leg raise on the left.

What is your diagnosis and how will you proceed in investigating the cause of his pain?

Plain Radiograph



Evaluate:
Alignment
Disc spaces
Osteophytes
Other bone abnormality

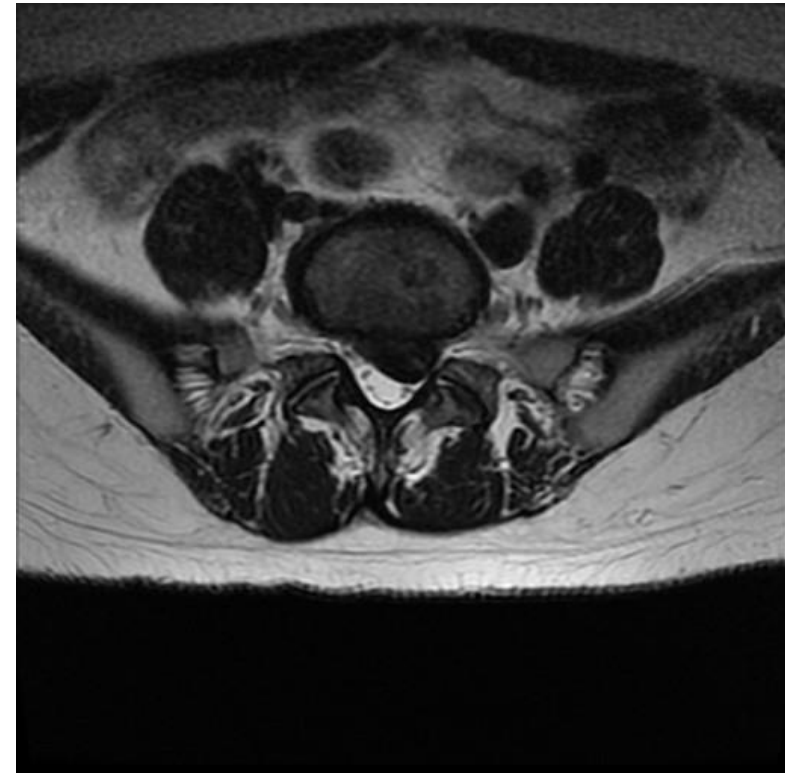
No information on disc pathology

Magnetic Resonance Imaging

Sagittal T2



Axial T2



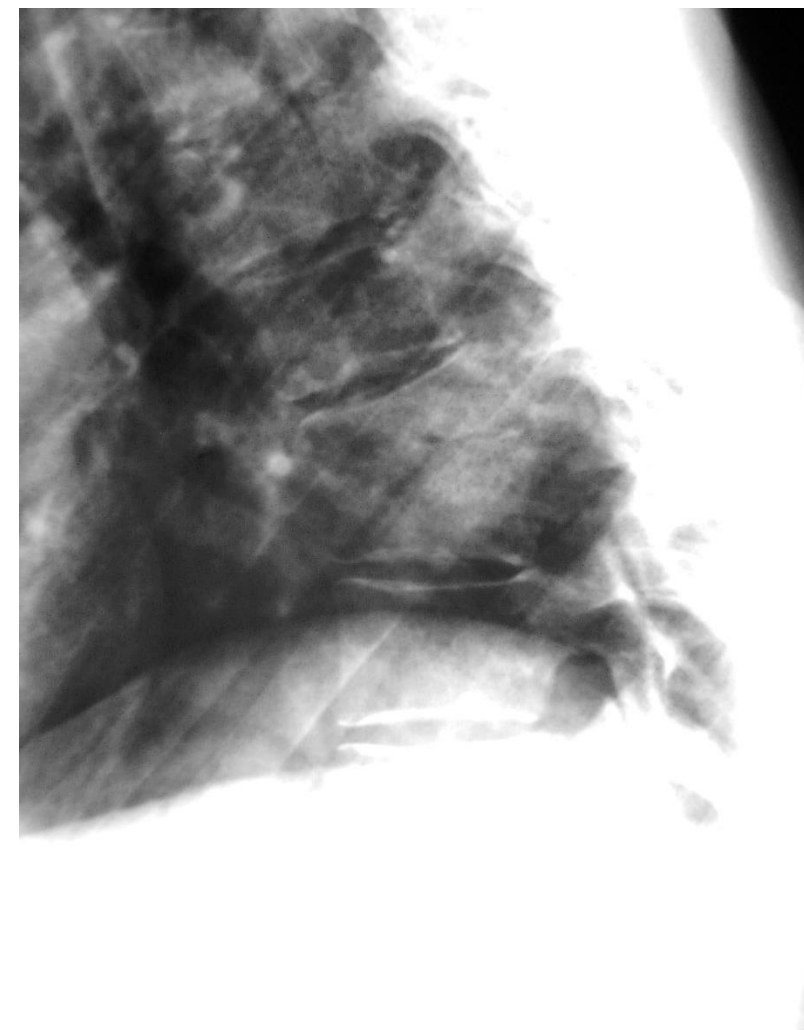
MRI is the choice of imaging to assess disc pathology: left L5/S1 disc protrusion

Case 8

- 65-year old female complained of an insidious onset of back pain with low grade fever and night sweats in the last 2 weeks.
- O/E: angular kyphus, tenderness in the mid-back, low grade fever.
- Elevated ESR.

What do you suspect and need to rule out?

T-spine x-ray

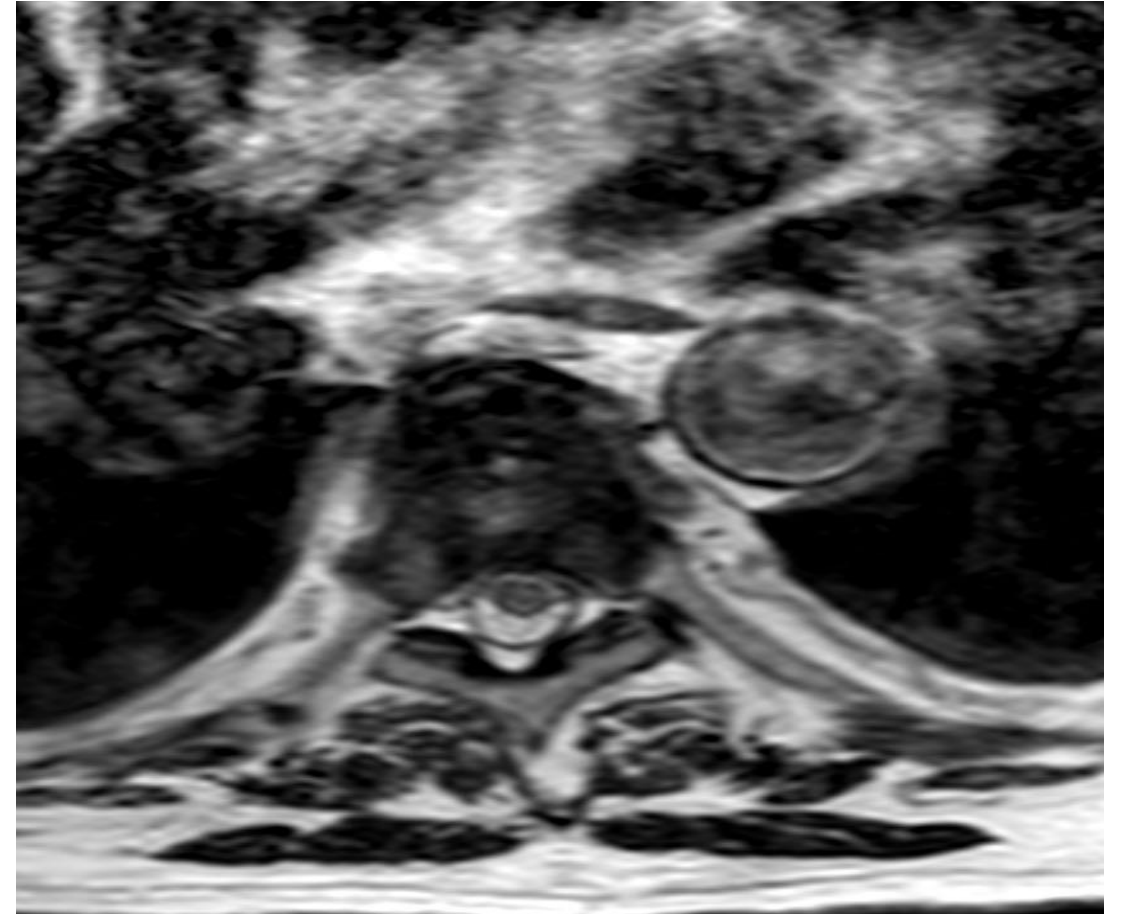


narrowing of
disc space

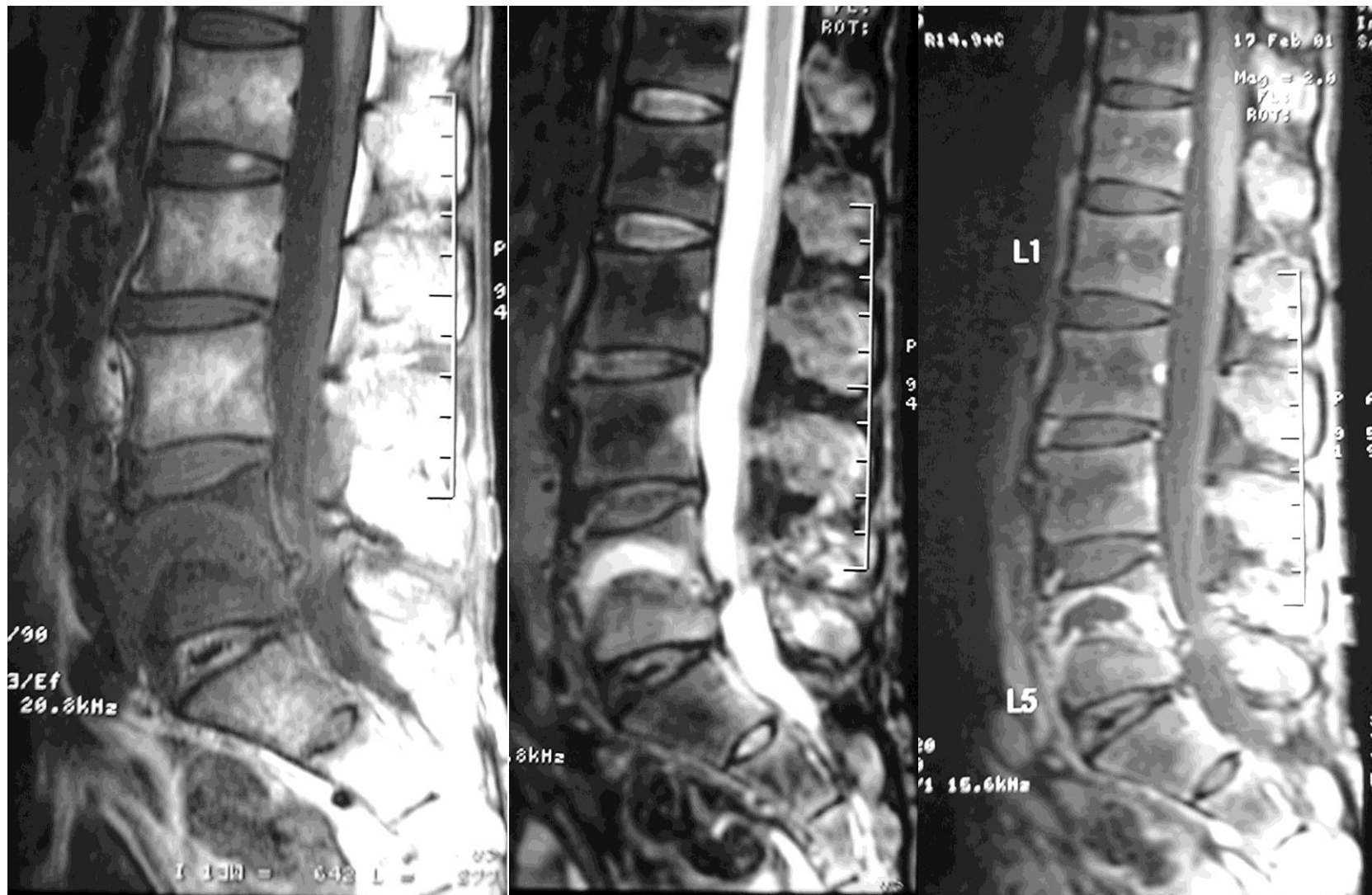
MRI

-disc-based pathology

-look for paravertebral soft tissue



MRI -discitis



T1

T2

T1 post-contrast

Learning Outcomes

- Develop a systematic approach to viewing MSK radiology
- Appreciate the features of common MSK pathologic conditions
- Understand the applications of various imaging techniques in MSK imaging.